

AI and ML for Architecture (M25RD204)

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Course Contents & Schedule

1. Python Programming + Relevant Libraries : 2.7.2026
2. Introduction to AI, Data Science and ML – Data Collection and Preprocessing : 2.7.2026
3. Supervised Learning Algorithms – Linear Regression for Space Planning : 2.7.2026
4. Unsupervised Learning – K - Means for Site Analysis : 2.7.2026
5. Neural Networks, Deep Learning : 6.7.2026
6. Computer Vision – CNNs, Image Classification, Stable Diffusion, Mid Journey : 6.7.2026
7. Generative Design with GANs : 7.7.2026
8. Generative AI – LLMs, VLMs, Multimodal AI Models : 7.7.2026
9. Generative AI Tools for Architecture : 7.7.2026

Course Contents

- 10. Natural Language Processing for Design Briefs – Text Analysis and Brief Processing : 8.7.2026
- 11. Reinforcement Learning for Space Utilization : 8.7.2026
- 12. Optimization Algorithms – Genetic Algorithm for Layout Planning : 9.7.2026
- 13. Building Performance Prediction – Energy Consumption ML Models : 13.7.2026
- 14. Smart Building Systems – IoT Integration with ML Planning : 13.7.2026
- 15. Advanced Applications - Custom AI Tool Development : 14.7.2026
- 16. AI Ethics in Architecture – Bias Detection and Mitigation : 14.7.2026
- 17. Industry Case Studies – Portfolio Compilation: 14.7.2026
- 18. Project Presentations – 15.7.2026

Role of Generative AI in Computational Design

Paradigm shift from strictly **parametric and rule-based** (using code to calculate exact coordinates, vectors, and panels), to **probabilistic, vision-based design**.

1. Hyper-Accelerated Concept Ideation (The Front-End): Mood board on steroids.

Type a prompt specifying materials, atmospheric conditions, and architectural typologies (e.g., "*A biophilic research center, rammed earth and curved glass, foggy morning, architectural photography*"), and receive high-fidelity conceptual imagery in seconds. You can test hundreds of aesthetic directions before writing a single line of code or clicking a mouse in Rhino.

Role of Generative AI in Computational Design

2. Image-to-Image (Img2Img) and Massing Transmutation : You can use Image to Image inputs to take a crude, blocky 3D massing model from Rhino and use AI to generate the architectural detailing, materiality, and lighting over that exact geometry.

- [Crude Rhino 3D Blocks] —> [AI Processing + Prompt] —> [Photo-Real Architectural Rendering]
- Automating the tedious task of texture mapping and scene rendering during early client presentations.

Role of Generative AI in Computational Design

3. Dematerializing the "Technical Wall"

- You don't need to be an expert in understanding vector math and syntax to see a complex form. Generative AI allows for **semantic design**—using natural language to describe spatial complexity.
- This democratizes the exploration of highly intricate, non-linear geometries (like Baroque detailing or complex organic lattices) for students who haven't yet mastered complex NURBS modelling in Rhino.

Role of Generative AI in Computational Design

4. Semantic vs. Explicit Geometry: Generative AI generates pixels, not data.

Attribute	Generative AI (e.g., Midjourney)	Computational Design (e.g., Rhino/Grasshopper)
Output Type	2D Pixel Grid (Images)	3D Mathematical Data (NURBS, Coordinates)
Precision	Probabilistic (Approximated, "Looks right")	Deterministic (Exact, "Is right")
Downstream Use	Client Inspiration, Conceptual Ideation	Fabrication, CNC Milling, Structural Analysis, BIM

Midjourney can dream up a breathtaking, highly complex structural column, but it cannot tell you how thick the steel must be, it cannot export a 3D file to a robotic arm, and it cannot calculate the surface area for cost estimation.

Using both Gen AI and Conventional Design Software in a modern Pipeline

- **Ideation:** Use **Midjourney** to rapidly discover a unique aesthetic language, material palette, or formal silhouette.
- **Translation:** Take that AI-generated inspiration image into **Rhino/Grasshopper**, using Python or visual scripting to decode the underlying geometric logic into actual 3D points and parametric vectors.
- **Execution:** Optimize that 3D model for structural engineering and digital fabrication.

AI is a Tool, Not a Replacement

- Gen AI does not replace architects and designers, it acts as a powerful front-end catalyst, while engines like Rhino 8 remain the mandatory backbone for turning those digital dreams into buildable reality.